SSc-MIM

Building the map, validating it, and characterising SSc endotypes

Construction · validation (gates + patient data) · endotype use case

Nathan Foulquier — LBAI, UMR 1227 Inserm · CHU de Brest

CONTEXT

The map in one screen

568
molecular species

308
mechanistic reactions

133
SSc-specific reactions

197
patient donors

A curated, SBGN/SBML disease map for dcSSc skin fibrosis.

Four modules — M1 type-I IFN · M2 TGF-β/fibrosis · M3 EndoMT/vasculopathy · M4 IL-6/Th2/B — draining into 4 phenotype sinks.

Topology is literature-curated; patient single-cell data is layered on to validate it and read it out.

This talk: how it is built, how it is validated (two ways), and how it can characterise patient endotypes.

PART I

Building the map

Literature-curated topology, assembled and annotated — not inferred from omics.

From literature to an integrated map

Scope — Reactome import — Harmonise & dedup — Integrate — SSc curation (gated) — Annotate PMID/ECO — QC

Backbone (blue)

Canonical pathways imported from Reactome (IFN-I, TGF-β/PDGF, NOTCH1, IL-6), harmonised to HGNC, de-duplicated, merged into one SBML map.

SSc-specific layer (coral)

133 hand-curated reactions Reactome cannot hold (autoantibodies, SSc cell states, crosstalk) — added only through gated discovery (Part II).

```
M1 IFN 19   M2 fibrosis 58
M3 EndoMT 27   M4 immune 21   xt 8
```

4 phenotype sinks: myofibroblast · ECM · vascular remodelling · ISG signature.

Rule: every entity reaches a sink in ≤6 steps (0 violations).

PART II

Validating the map

Two complementary axes: (A) gated against fabrication, (B) grounded in real patient data.

VALIDATION

Two complementary axes

A · Is every edge real?

Edge-discovery protocol: nothing enters the curated map speculatively.

Five automated gates (G0–G4), G2 = verbatim-quote grounding.

Contradiction detector + advisory AI verdicts + citation-integrity sweep.

Binding human ratification by the expert/co-author.

B · Does it hold in patients?

Four single-cell datasets (197 donors) re-analysed from raw counts.

Per-donor pseudobulk → mixed-effects DEG → map coverage.

AUCell → per-patient module activation profile.

Result: 81.3% coverage; type-I IFN validated in SSc skin.

Gate strategy — the edge-discovery protocol

Governing rule: the curated map is never written speculatively.

1. **Stage** — candidate edge in a staging file — with a verbatim sentence from the source paper
2. **Fetch** — cache the source text (Europe PMC OA / abstract)
3. **Gate** — run the five G0–G4 gates → PASS / FLAG / REJECT
4. **Ratify** — a human sets decision = promote (default = held)
5. **Promote** — only PASS + promote enter the map — tagged, reversible

The five gates (G0–G4)

Gate	Check	On failure
G0 schema	required fields present; type in controlled vocabulary	REJECT
G1 HGNC	every gene entity is an official HGNC symbol	REJECT (alias→FLAG)
G2 grounding	supporting quote is a verbatim substring of the source text	REJECT — keystone
G3 novelty	gene pair not already an SSc reaction; not a forward Reactome pair	REJECT (dup) / FLAG
G4 evidence	numeric PMID present; corpus text available (SSc context)	FLAG

If the quote is not literally in the paper, the edge is rejected.

Self-test: a fabricated negative-control edge (FOXP3→COL1A1, invented quote + decoy PMID) must REJECT on every run.

Then: contradiction detector · AI verdicts (128 validate / 5 caution) · citation sweep (28 wrong PMIDs fixed, re-verified vs PubMed) · human ratification.

Grounding in patient data — the datasets

197 donors (121 SSc / 76 HC) across skin, lung and blood — re-analysed from raw GEO files; citations verified on PubMed.

GEO	Reference (verified)	Tissue	Donors (SSc/HC)	Cells
GSE195452	Gur C et al. Cell 2022;185:1373	Skin (multiome)	154 (97 / 57)	100 538
GSE138669	Tabib T et al. Nat Commun 2021;12:4384	Skin	22 (12 / 10)	64 211
GSE128169	Morse C et al. Eur Respir J 2019;54:1802441	Lung (ILD)	13 (8 / 5)	67 516
GSE210395	GEO — no linked publication	PBMC	8 (4 / 4)	34 619

Skin cohorts give the fibroblast-subset resolution that M2/M3 model; lung (SSc-ILD) + PBMC extend beyond skin.

Raw archives SHA-256-pinned in data/MIRROR.md for a reproducible input envelope (Zenodo mirror).

Per-donor pseudobulk — the right unit of replication

Tens of thousands of cells, but only 4–13 patients per dataset. The question is about patients, not cells.

The problem: pseudoreplication

Cells from one patient are not independent (same genotype, biopsy, batch).

Testing per-cell treats them as thousands of samples → inflated power, false positives.

The fix

Rebuild one bulk-like profile per patient → the unit of replication becomes the patient.

```
group cells by (donor × cell_type)
row[gene] = Σ raw counts over cells
```

→ one pseudobulk count row per (donor, cell type).

```
NB GLM (pydeseq2) ~ condition
+ Benjamini-Hochberg FDR (q=0.05)
```

Raw counts (not log) — the NB GLM models depth & overdispersion itself.

Trade-off: fewer but robust hits; a valid, reproducible inference. The same pseudobulks feed AUCell.

AUCell — label-free module activation scoring

"Are a module's genes concentrated among this patient's most highly-expressed genes?" If yes → the module is on.

Why — no double-dipping

Scores each donor against the pre-specified module gene set (from the curated map), blind to the SSc/HC label.

Avoids the circularity of weighting by the same SSc-vs-HC contrast you then test.

Rank-based → robust

Depends only on gene order, not absolute values → insensitive to scale, depth, normalisation.

Rank all genes by expression (rank 1 = highest).

Walk the top T (top ~5%); recovery curve R(r) = # module genes with rank ≤ r.

`AUC = Σ R(r),  normalised to [0,1]`

High AUC → module genes cluster at the top → module active.

Output: a 4-module activation vector per patient.

This per-patient vector is exactly what powers the endotype use case (Part III).

What the data shows

81.3%

of detectable map species seen in patient data

Coverage by module

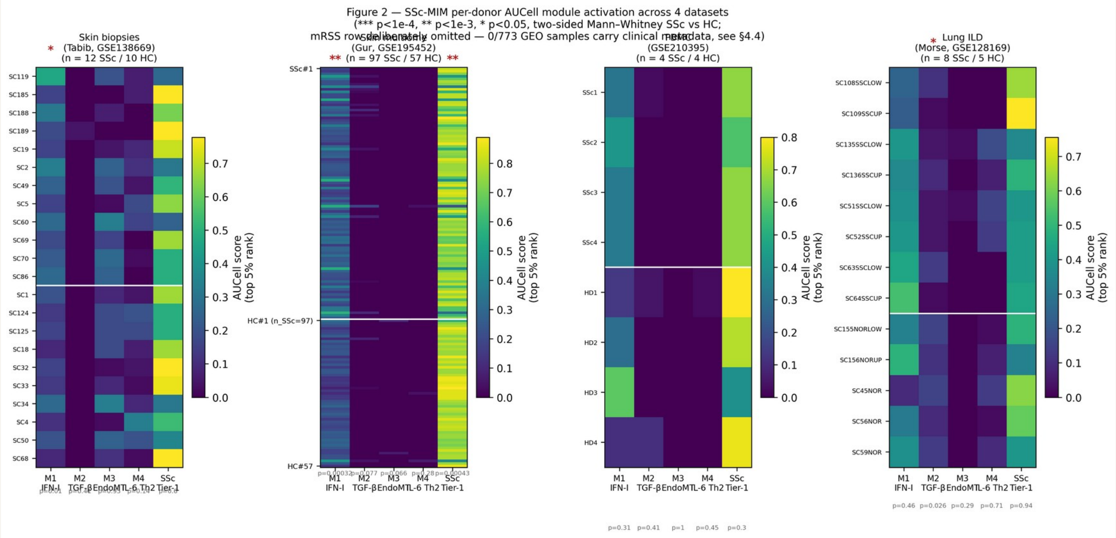
M1 84% · M2 88% · M3 75% · M4 71% — the curated biology is largely measurable.

Biological validation

Type-I IFN (M1) significantly elevated in SSc skin: Gur p=3.2e-4, Tabib p=0.011.

Annotation validated

CellTypist agreement κ=0.92, ARI=0.70 vs marker labels.



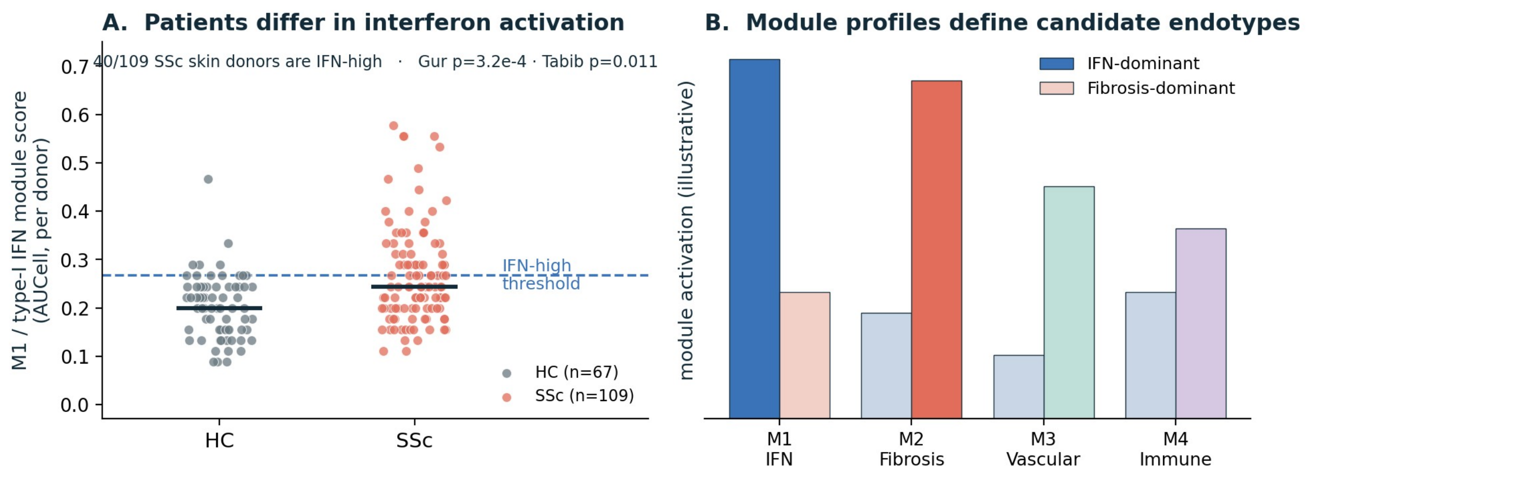
Per-donor AUCell module scores, SSc vs HC.

PART III

Toward endotypes

Using the validated map to characterise patient subgroups — mechanistically.

Patients already differ in module activation



Each patient → a 4-module activation vector. Distinct profiles → mechanism-matched endotypes (IFN-high → JAK/IFN axis · fibrosis-high → TGF-β/CTGF axis), recapitulating known SSc intrinsic subsets.

Each patient = a 4-module activation vector (Part II-B).

IFN activation is not uniform: 40/109 SSc skin donors are IFN-high, others are not (reproducible Gur + Tabib).

Distinct module profiles → candidate mechanism-matched endotypes (IFN-dominant vs fibrosis-dominant).

Recapitulates the known SSc intrinsic subsets (inflammatory ↔ M1/M4, fibroproliferative ↔ M2).

USE CASE

You arrive with 3 patient clusters — what the map gives you

Project any patient grouping onto the map — same pipeline (AUCell + per-cluster DEG + MINERVA overlay), your labels.

YOUR CLUSTERS

Cluster 1

Cluster 2

Cluster 3

→

AUCell — score each cluster on the 4 modules (M1/M2/M3/M4 + Tier-1).

Per-cluster differential expression mapped onto map species → which reactions light up.

MINERVA overlay — colour the map by each cluster (any label, no new method).

WHAT YOU GET BACK — per cluster

Module fingerprint

a 4-module activation vector — e.g. cluster 2 IFN-high, cluster 3 fibrosis-dominant.

Mechanistic hypotheses

the lit-up sub-circuits that distinguish each cluster — e.g. STING→IRF3 vs TGF-β→SMAD3→collagen.

Candidate targets

active druggable hubs per cluster → a cluster-matched therapy hypothesis to test.

In one line

A curated map, gated against fabrication, validated in patients — and ready to type them

Literature backbone + 133 gated SSc-specific reactions → one SBML map (568 sp / 308 rx).

G0–G4 + G2 grounding + human ratification: no edge without a verbatim source.

4 datasets, 197 donors; per-donor pseudobulk + AUCell; 81.3% coverage; IFN validated in SSc skin.

Project your patient clusters → per-cluster module fingerprint, mechanistic hypotheses, candidate targets.